

Haptics System

1.0.0

Generated by Doxygen 1.9.5

1 Namespace Index	1
1.1 Package List	1
2 Hierarchical Index	3
2.1 Class Hierarchy	3
3 Class Index	5
3.1 Class List	5
4 Namespace Documentation	7
4.1 HapticsSystem Namespace Reference	7
4.2 HapticsSystem.Structures Namespace Reference	7
5 Class Documentation	9
5.1 HapticsSystem.Structures.FloatMinMax Struct Reference	9
5.2 HapticsSystem.IHapticsDevice Class Reference	9
5.2.1 Detailed Description	10
5.2.2 Member Function Documentation	10
5.2.2.1 PlayHapticImpulse()	10
5.2.2.2 TryHapticImpulse()	10
5.3 HapticsSystem.PlayHapticsOnCollision Class Reference	11
5.3.1 Detailed Description	11
5.3.2 Member Function Documentation	11
5.3.2.1 PlayHaptics()	12
Index	15

Chapter 1

Namespace Index

1.1 Package List

Here are the packages with brief descriptions (if available):

HapticsSystem		7
HapticsSystem.Structures		7

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

HapticsSystem.Structures.FloatMinMax	9
MonoBehaviour	
HapticsSystem.IHapticsDevice	9
HapticsSystem.PlayHapticsOnCollision	11

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

HapticsSystem.Structures.FloatMinMax	9
HapticsSystem.IHapticsDevice	
The haptics driver is responsible for actually playing haptics for an input device.	9
HapticsSystem.PlayHapticsOnCollision	
A component that detects collisions of a modifiable severity and plays the configured haptics.	11

Chapter 4

Namespace Documentation

4.1 HapticsSystem Namespace Reference

Classes

- class [IHapticsDevice](#)
The haptics driver is responsible for actually playing haptics for an input device.
- class [PlayHapticsOnCollision](#)
A component that detects collisions of a modifiable severity and plays the configured haptics.

4.2 HapticsSystem.Structures Namespace Reference

Classes

- struct [FloatMinMax](#)

Chapter 5

Class Documentation

5.1 HapticsSystem.Structures.FloatMinMax Struct Reference

Public Attributes

- float **minimum**
The minimum value.
- float **maximum**
The maximum value.

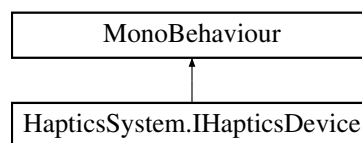
The documentation for this struct was generated from the following file:

- FloatMinMax.cs

5.2 HapticsSystem.IHapticsDevice Class Reference

The haptics driver is responsible for actually playing haptics for an input device.

Inheritance diagram for HapticsSystem.IHapticsDevice:



Public Member Functions

- void [TryHapticImpulse](#) (float pDuration, float pAmplitude, float pFrequency)
Attempts to play a haptic pulse with the specified parameters on all devices registered with this haptics device. Note that haptics will only play while the haptics device component is enabled.
- abstract void [PlayHapticImpulse](#) (float pDuration, float pAmplitude, float pFrequency)
Attempts to play a haptic pulse with the specified parameters on all devices registered with this haptics device. NOTE: This differs from 'TryHapticImpulse' in that it will play the impulse regardless of whether or not the component is enabled.
- abstract void **StopHaptics** ()
Stops haptics on all registered haptic devices.

5.2.1 Detailed Description

The haptics driver is responsible for actually playing haptics for an input device.

Author: Mathew Aloisio

5.2.2 Member Function Documentation

5.2.2.1 PlayHapticImpulse()

```
abstract void HapticsSystem.IHapticsDevice.PlayHapticImpulse (
    float pDuration,
    float pAmplitude,
    float pFrequency ) [pure virtual]
```

Attempts to play a haptic pulse with the specified parameters on all devices registered with this haptics device. NOTE: This differs from 'TryHapticImpulse' in that it will play the impulse regardless of whether or not the component is enabled.

Parameters

<i>pDuration</i>	
<i>pAmplitude</i>	
<i>pFrequency</i>	

5.2.2.2 TryHapticImpulse()

```
void HapticsSystem.IHapticsDevice.TryHapticImpulse (
    float pDuration,
    float pAmplitude,
    float pFrequency )
```

Attempts to play a haptic pulse with the specified parameters on all devices registered with this haptics device. Note that haptics will only play while the haptics device component is enabled.

Parameters

<i>pDuration</i>	
<i>pAmplitude</i>	
<i>pFrequency</i>	

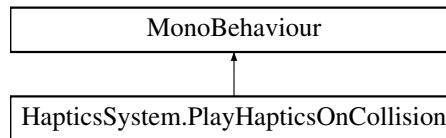
The documentation for this class was generated from the following file:

- IHapticsDevice.cs

5.3 HapticsSystem.PlayHapticsOnCollision Class Reference

A component that detects collisions of a modifiable severity and plays the configured haptics.

Inheritance diagram for HapticsSystem.PlayHapticsOnCollision:



Public Member Functions

- void [PlayHaptics](#) (float pNormalizedSeverity)
Plays haptics for a collision with a normalized severity (0 to 1) of pNormalizedSeverity. Higher severity increases haptic amplitude within the specified range. Higher severity increases haptic frequency within the specified range.

Public Attributes

- [FloatMinMax](#) **collisionVelocityThreshold**
The range of collision velocity thresholds. Haptics will be played on collisions with a relative velocity square magnitude above collisionVelocityThreshold.minimum
- List< [IHapticsDevice](#) > **hapticsDevices** = new List<[IHapticsDevice](#)>()
A reference to the HapticsManager associated with this component.
- float **hapticsDuration** = 0.3f
The number of seconds to play haptics for on a collision.
- [FloatMinMax](#) **hapticsAmplitude**
The haptics amplitude range to play for a collision.
- [FloatMinMax](#) **hapticsFrequency**
The haptics frequency range to play for a collision.
- float **hapticsRepeatDelay** = 0.3f
The minimum number of seconds before haptics can be played again after being played.

Properties

- float **LastHapticsPlayRealTime** [get]
Returns the last Time.realTimeSinceStartup this component played haptics.

5.3.1 Detailed Description

A component that detects collisions of a modifiable severity and plays the configured haptics.

Author: Intuitive Gaming Solutions

5.3.2 Member Function Documentation

5.3.2.1 PlayHaptics()

```
void HapticsSystem.PlayHapticsOnCollision.PlayHaptics (
    float pNormalizedSeverity )
```

Plays haptics for a collision with a normalized severity (0 to 1) of *pNormalizedSeverity*. Higher severity increases haptic amplitude within the specified range. Higher severity increases haptic frequency within the specified range.

Parameters

<i>pNormalizedSeverity</i>	
----------------------------	--

The documentation for this class was generated from the following file:

- PlayHapticsOnCollision.cs

Index

- HapticsSystem, [7](#)
- HapticsSystem.IHapticsDevice, [9](#)
 - PlayHapticImpulse, [10](#)
 - TryHapticImpulse, [10](#)
- HapticsSystem.PlayHapticsOnCollision, [11](#)
 - PlayHaptics, [11](#)
- HapticsSystem.Structures, [7](#)
- HapticsSystem.Structures.FloatMinMax, [9](#)
- PlayHapticImpulse
 - HapticsSystem.IHapticsDevice, [10](#)
- PlayHaptics
 - HapticsSystem.PlayHapticsOnCollision, [11](#)
- TryHapticImpulse
 - HapticsSystem.IHapticsDevice, [10](#)